

Online Retails Sales: Customer Segmentation and Profiling

Probability and Statistics for Artificial Intelligence (AAI-500-IN2)



AA500:
Group 9



Meet the team

Nikhil Bembi

Team Lead

Led the work on the k-means and hierarchical clustering model.

Ms. Nehpreet Kaur

Team Member

Led the documentation and preparation of final submissions.

Tamayi Mlanda

Team Member

Led the work on the EDA and RFM model.

Dataset Selection

We chose to work with the Online Retail sales dataset which is found at:

UCI Repository Dataset:

<https://archive.ics.uci.edu/dataset/352/online+retail>

Project GitHub:

https://github.com/nbemb/AA_500_USD_Group9_CustomerSegmentation/tree/dev





Problem

Dataset selection

We chose the Online Retail dataset for a UK based online sales company. During our EDA, we formulated a number of research questions and possible areas for investigation.

Market gap

We have noticed that many customers are not returning to buy from the shop. We would like to identify the different groups of customers, understand their buying behavior and develop mitigation strategies to strengthen the relationships with these customers.

Customers

Customers are coming from different countries, especially the United Kingdom.

Financials

The average Customer spends an average of GBP 20.03 per order

Solution



Understand the Data

Perform EDA to understand the dataset and check for data quality and statistical metrics and properties of the data.



Customer Segments

Perform a RFM analysis to understand the different customer groups.



Data Modelling

Develop statistical models that help to understand the data and customer segments better.



Advise the Business

Provide recommendations and solutions to the business for strategies they can follow to improve sales and customer retention.

Recency, Frequency and Monetary Value (RFM) Analysis

Quartiles

Quartile-based scoring (1 - 4) will be used for simplicity and interpretability.

Distribution of RFM Scores

Understand the proportion of high-value vs. low-value customers, which is critical for marketing strategies

Recency vs Monetary Analysis

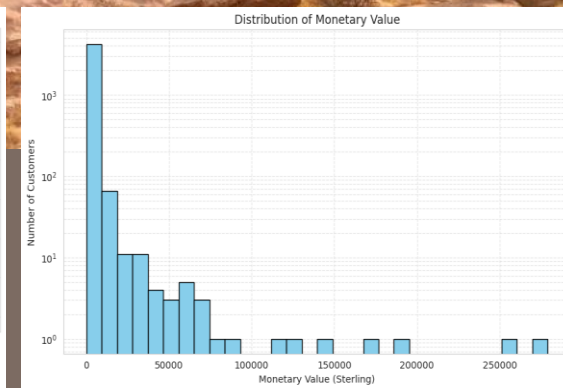
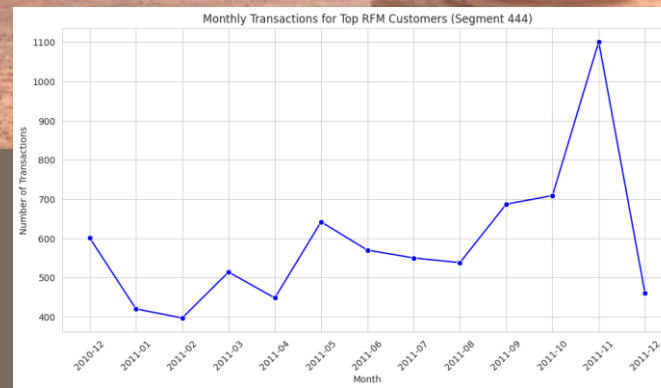
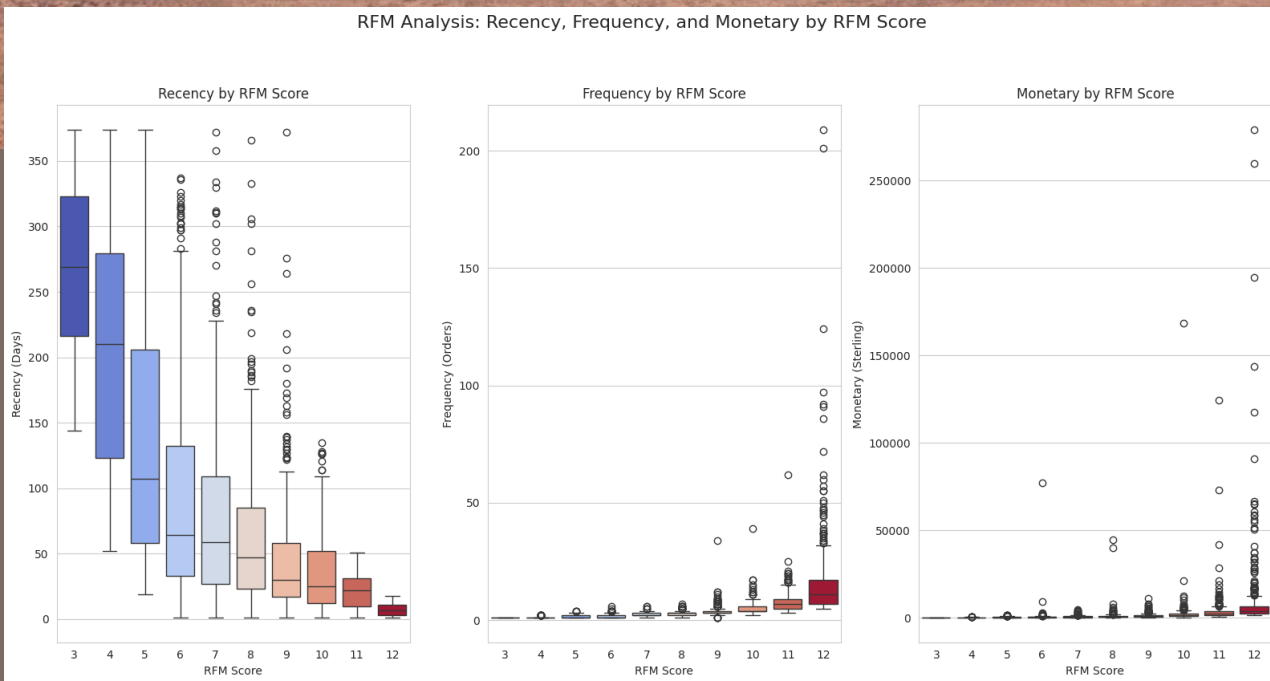
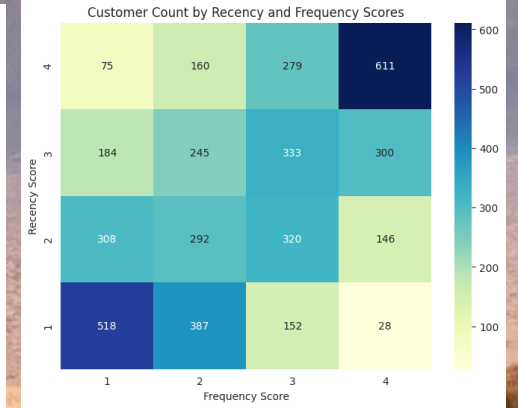
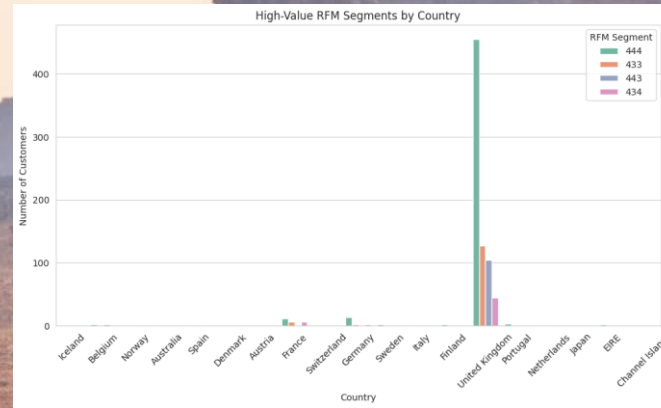
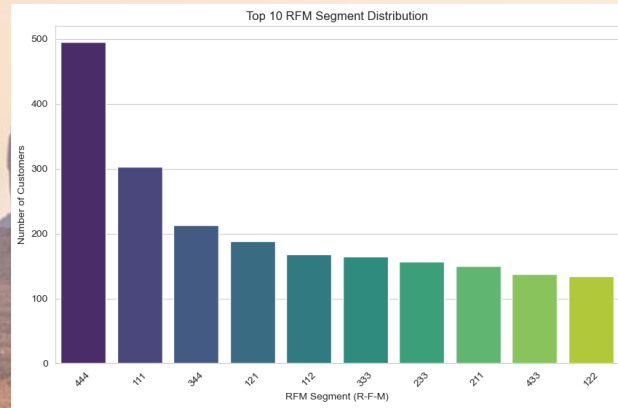
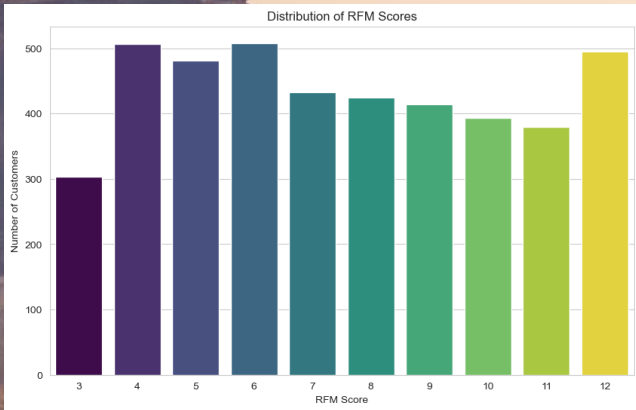
Insights into customer behaviour based on Recency (days since last purchase) and Monetary value (total spend).

Customer Spending Patterns

Understand the behaviour our customers spending patterns.



RFM Analysis Outputs



A comprehensive understanding of the customer behaviour from multiple perspectives. From this, we can start deriving some recommendations and build further statistical models.

Business Strategy & Recommendations

Growth

Opportunity to identify customers that should be prioritized for retention through loyalty programs or exclusive offers.

Re-engagement

Re-engagement campaigns targeting these customers could prevent churn. Upselling or increasing purchase frequency could move them toward higher scores.

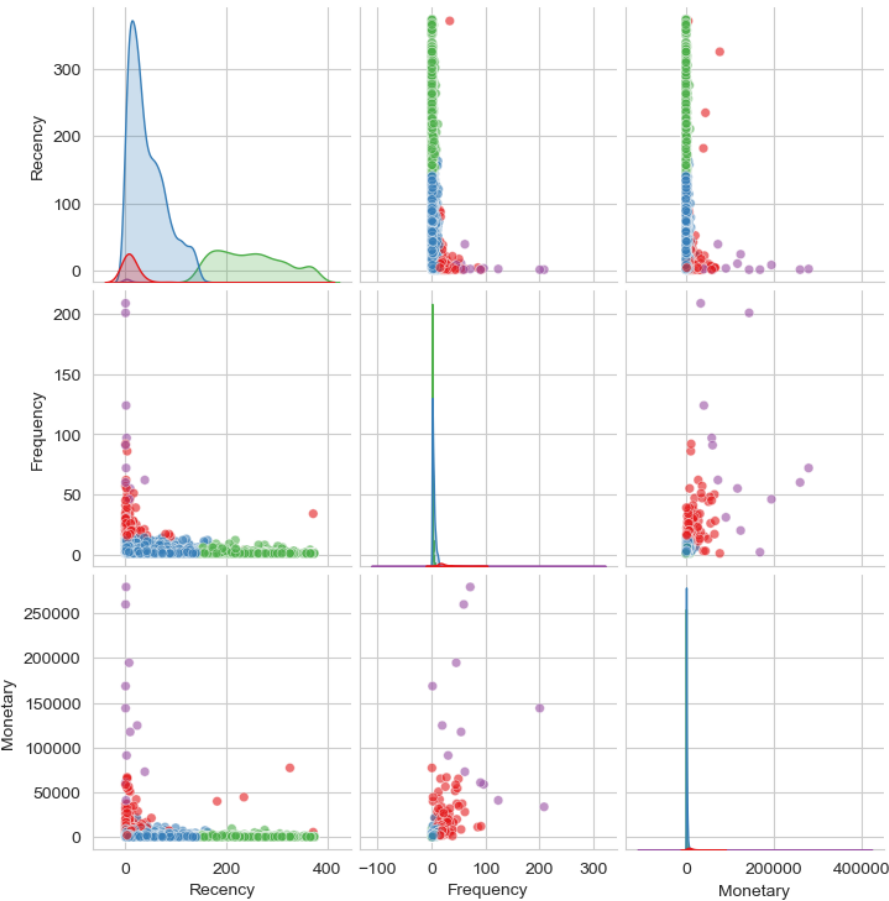
Deep-Dive

More detailed analysis required. This is the middle ground of customers with moderate recency, frequency, or monetary value.



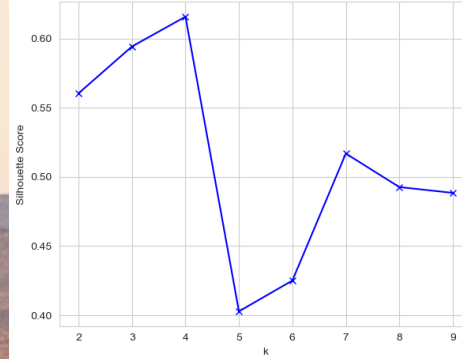
K-Means Clustering Outputs

Pairplot of RFM by Cluster (KMeans)

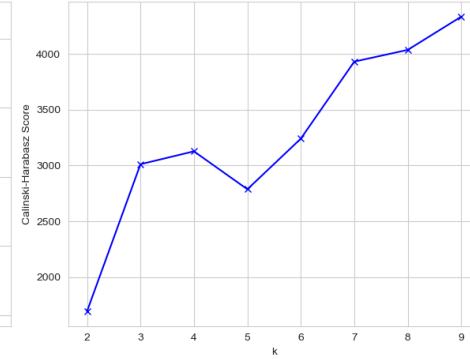


Cluster
 ● 0
 ● 3
 ● 1
 ● 2

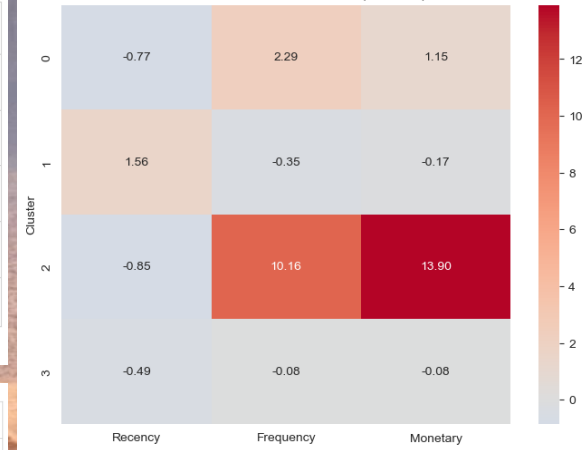
Silhouette Score vs. k



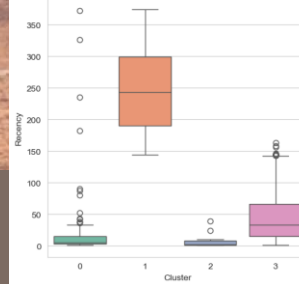
Calinski-Harabasz Score vs. k



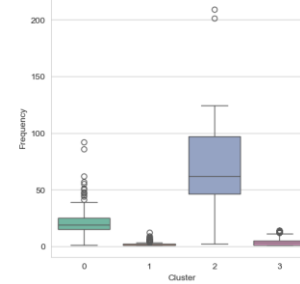
Standardized Cluster Centroids (KMeans)



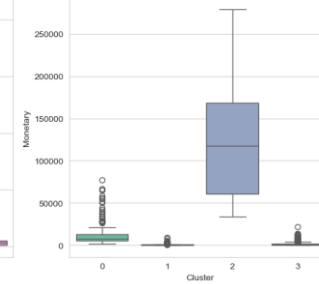
Recency Distribution by Cluster



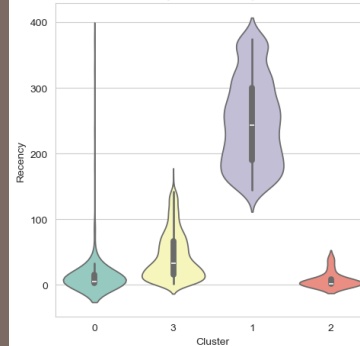
Frequency Distribution by Cluster



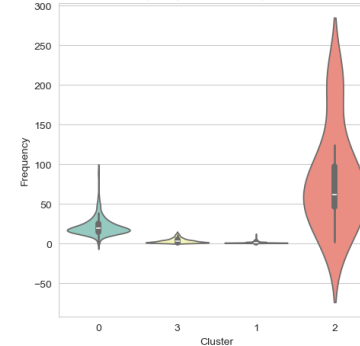
Monetary Distribution by Cluster



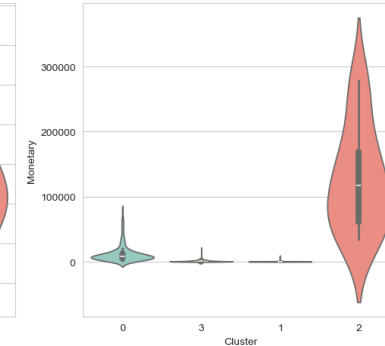
Recency Distribution by Cluster



Frequency Distribution by Cluster

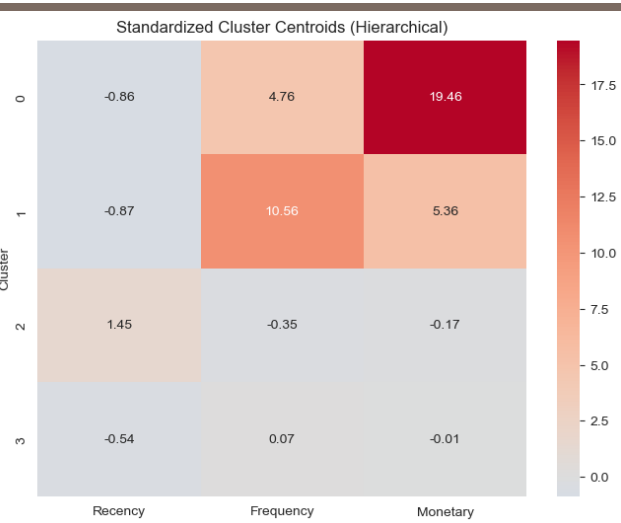
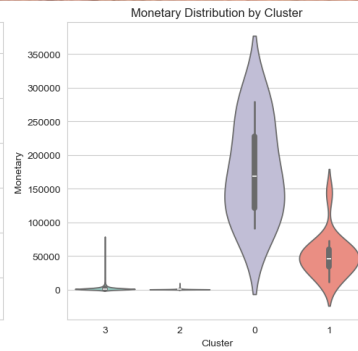
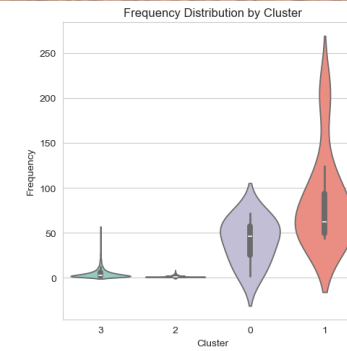
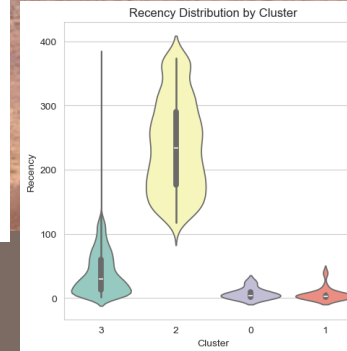
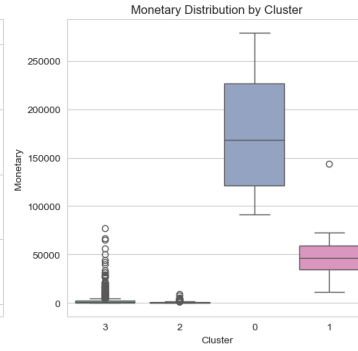
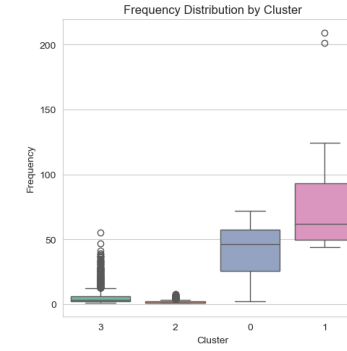
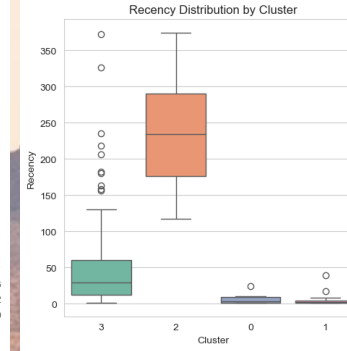
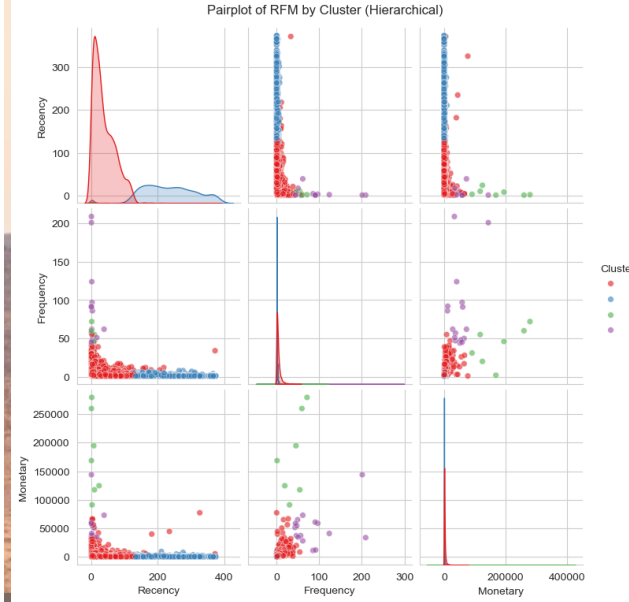
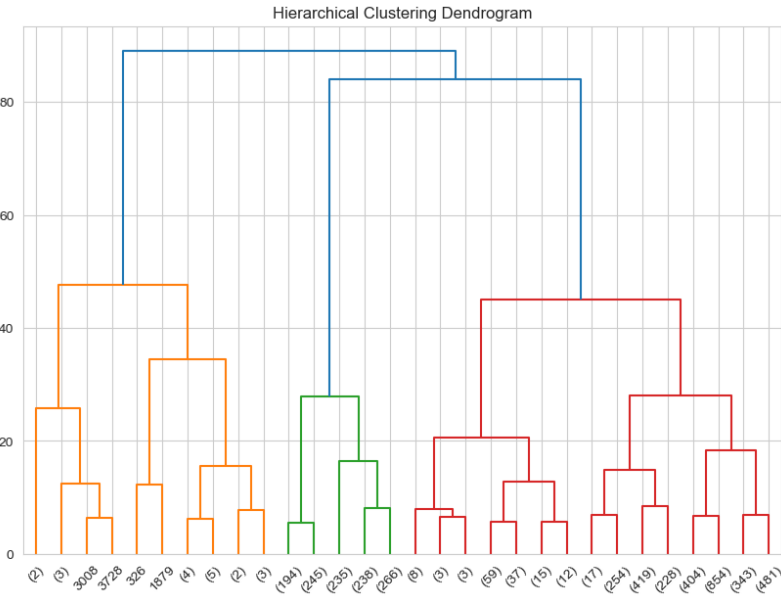


Monetary Distribution by Cluster



A comprehensive understanding of the customer behavior from multiple perspectives. From this, we can start deriving some recommendations and build further statistical models.

Hierarchical Clustering Outputs



A comprehensive understanding of the customer behavior from multiple perspectives.

From this, we can start deriving some recommendations and build further statistical models.

Findings

Through this analysis, we have identified some customer segments and the business strategies that we recommend for the business.

High-Value (e.g., Cluster 0): Bar plot/heatmap show low Recency, high Monetary → Offer VIP discounts (e.g., France).

At-Risk (e.g., Cluster 1): High Recency in scatters/box plots → Re-engagement emails (e.g., €5 vouchers in Norway).

Loyal (e.g., Cluster 2): Moderate RFM in bar plot, wide violin plots → Upsell bundles (e.g., Germany).

Emerging (e.g., Cluster 3): Moderate bars | Box plots in → Onboarding offers (e.g., free shipping).



Thank you

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